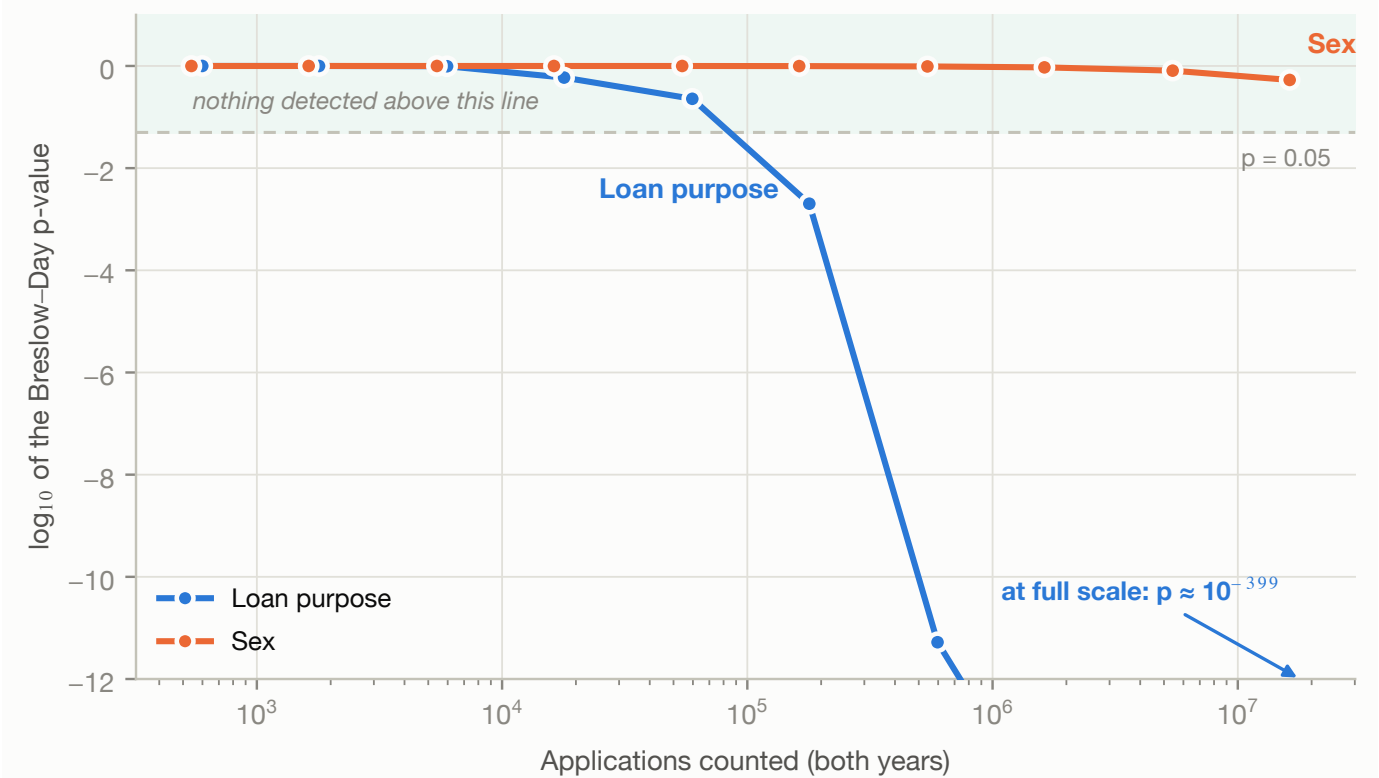


Significant by Volume

A homogeneity test on 17.9 million mortgage applications returns $p \approx 10^{-399}$.
The same data, counted at one thousandth of the scale, returns $p = 0.73$.
The segments did not change. The verdict did.



0 of 56 Simpson reversals — the warned-of failure	53 of 56 pairs where homogeneity is rejected	2.19× / 1.00× heterogeneity of the widest and narrowest axis
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Significant by Volume: What a Homogeneity Test on Approval Rates Measures, HMDA 2018–2025

The pooled approval rate is not undone by Simpson's paradox — that does not happen here at all. It is undone by heterogeneity, and the standard test for heterogeneity mostly reports how many applications were counted.

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Working paper · September 2026 · Data and code: github.com/cperezrisquet/approval-rate-reversal

ABSTRACT

Aggregate approval rates are standing indicators in credit operations, and the standard caution against them invokes Simpson's paradox. Across 8.6 to 9.3 million annual mortgage applications reported under the Home Mortgage Disclosure Act, eight segmentation axes and seven consecutive year-pairs — 56 tests in all — that paradox never occurs. The composition of demand moves the pooled rate by less than 0.5 pp in 49 of the 56 pairs. What does occur is heterogeneity: the Breslow–Day test rejects a common odds ratio in 53 of 56 pairs, 47 of them at $p < 10^{-10}$. That result is close to uninformative, because the same 2024–2025 loan-purpose data counted at one thousandth of scale returns $p = 0.73$ with the effect size unchanged. Whether a portfolio's segments "differ significantly" is therefore largely a statement about its volume. Reported as magnitude instead, the eight axes span $1.00\times$ to $2.19\times$ in the ratio of largest to smallest stratum odds ratio, and the ordering is not the intuitive one. The paper argues for publishing the spread rather than the test.

1. Introduction

An approval rate is the most portable indicator in consumer credit. It is one division, it is comparable across months and institutions, and it appears on nearly every operational dashboard. The standard caution against reading it in the aggregate is Simpson's paradox: a rate computed over a pooled population can move against every stratum that composes it. Yule (1903) described the reversal, Simpson (1951) formalised it, Bickel et al. (1975) found it in admissions data, and it has since become the canonical warning in the analyst's training.

This paper reports that the warning is misdirected, and that a more common failure sits behind it. In 56 tests over public mortgage-application data the reversal never appears. The pooled rate is nonetheless a poor summary of its strata, for a different reason — the strata do not share a common effect — and the test that is supposed to establish this cannot be read at face value, because at these sample sizes it rejects almost everything.

The contribution is not the observation that large samples produce small p-values, which is old (Berkson, 1938; Lin et al., 2013). It is the measurement: how much of a reported approval-rate movement is composition (almost none), how far the homogeneity verdict moves when only the sample size changes (from $p = 0.73$ to $p \approx 10^{-399}$ on identical proportions), and how much heterogeneity each candidate segmentation axis actually carries once magnitude replaces significance.

2. Data and method

All figures come from filings under the Home Mortgage Disclosure Act (HMDA), retrieved through the public Data Browser API of the Federal Financial Institutions Examination Council. No key or registration is required. The unit of

observation is a loan application; this paper uses the aggregated cell counts the API returns, which are sufficient for every quantity reported here.

Two properties of that endpoint shape the design. It does not serve a national aggregate, so a geographic filter is mandatory; the 50 states and the District of Columbia are requested, and that set defines the population of the paper. Territories are excluded. A multi-valued filter also acts as a grouping axis, so requesting the six loan-purpose codes returns the purpose-by-outcome cross-tabulation directly.

2.1 The approval rate is a definition, not a field

HMDA reports `action_taken`, not an approval rate. The rate used throughout is

1. numerator: code 1 (loan originated) plus code 2 (application approved but not accepted by the applicant) — in both the lender approved;
2. denominator: codes 1, 2 and 3 (application denied);
3. excluded: code 4 (withdrawn by the applicant), code 5 (file closed for incompleteness) and code 6 (purchased loan).

Codes 4 to 6 are excluded because the rate should measure lender decisions, and a withdrawal is not one. Including withdrawals is common in practice and changes the level; section 8 reports the consequence.

2.2 Estimands

For strata i with weights w_i and stratum rates r_i , the pooled rate is $\Sigma w_i r_i$. A change in it decomposes exactly — not approximately — into a within-stratum and a composition term,

$$\Sigma w_1 r_1 - \Sigma w_0 r_0 = \Sigma \bar{w} (r_1 - r_0) + \Sigma \bar{r} (w_1 - w_0)$$

with $\bar{w}$ and $\bar{r}$ the two-period averages. A Simpson reversal is recorded only when the pooled rate moves in one direction and *every* comparable stratum moves in the other. Unanimity is required deliberately: if one stratum dissents the phenomenon is an ordinary composition effect, and conflating the two is what makes the paradox seem ubiquitous. Heterogeneity across strata is tested with Breslow–Day (Breslow & Day, 1980) and summarised by magnitude as the ratio of the largest to the smallest stratum odds ratio. Mantel–Haenszel (1959) supplies a common odds ratio for comparison, and direct standardisation supplies the rate under a fixed composition.

2.3 Pre-registration

STATED BEFORE THE DATA WERE EXAMINED

The segmentation was fixed *ex ante* on a mechanism: refinancing volume tracks interest rates far more sharply than purchase volume, and refinancings carry a different approval rate, so a shift toward refinancing should drag the pooled rate down while each purpose improves. The prediction was a negative composition term larger in magnitude than the within-stratum term. **It was false** — the composition term is -0.07 pp against a within-stratum term of $+1.74$ pp. The record is in the repository, and the sweep in section 6 reports the whole distribution across axes rather than the largest value in it.

3. Result 1: the reversal does not occur

Across eight segmentation axes and seven consecutive year-pairs from 2018 to 2025, no pair produces a unanimous reversal. The composition term is smaller than 0.5 pp in absolute value in 49 of the 56 pairs, with a median of 0.07 pp and a maximum of 2.18 pp, against pooled movements ranging from 0.03 pp to 6.5 pp. In these data the pooled approval rate moves because decisions within segments changed, not because the mix of applicants changed.

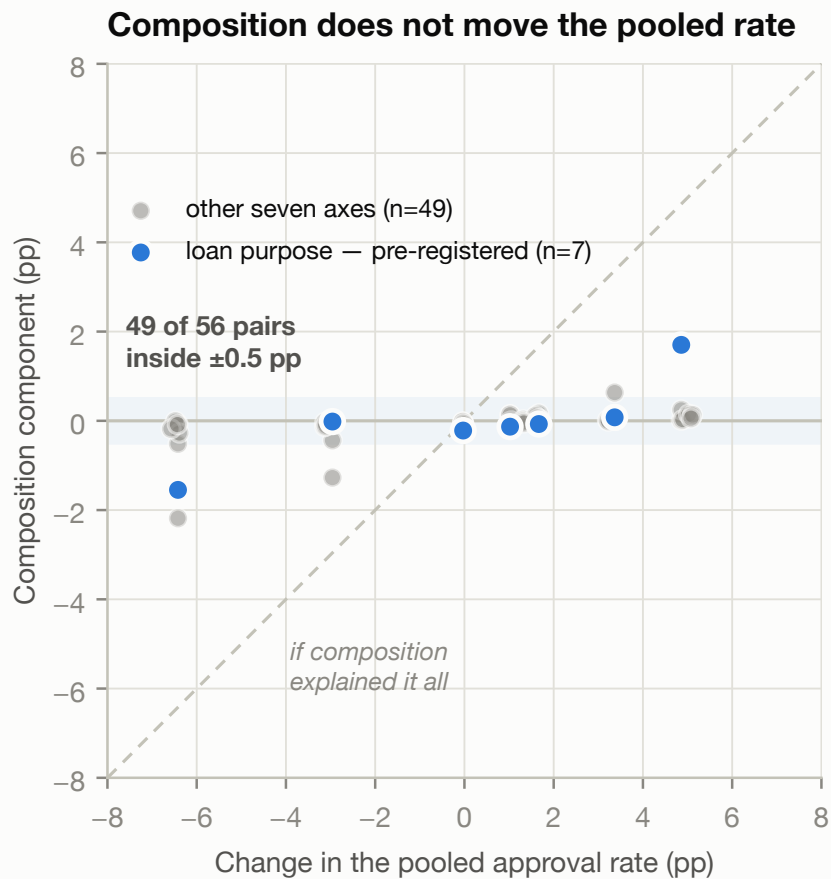


Figure 1. Composition against the aggregate, all 56 segmentation-by-year-pair tests. If a change in the mix of applicants accounted for the movement in the pooled approval rate, the points would lie on the dashed diagonal. They lie on the horizontal instead: 49 of 56 fall within ± 0.5 pp of zero. Loan purpose is highlighted because it was the pre-registered axis.

The closest case is loan purpose between 2020 and 2021, where five of six strata move against the aggregate carrying 62.6 % of the weight. The pooled change there is -0.03 pp, indistinguishable from zero; reporting it as a sign reversal would be an artefact of dividing by a quantity that is not meaningfully different from nothing. It is counted as no reversal.

4. Result 2: homogeneity is rejected almost everywhere

The pooled rate can fail as a summary without any reversal, and here it does. Breslow–Day rejects a common odds ratio in 53 of the 56 pairs, 47 of them at $p < 10^{-10}$. For 2024–2025 by loan purpose the statistic is 1855.9 on 5 degrees of freedom, which corresponds to $p \approx 10^{-399}$.

That last number is reported to make a point about its own worthlessness. Each pair carries between 13.3 and 36.7 million applications. At those sizes the test has power against differences far too small to matter operationally, and a p-value near zero cannot distinguish an axis that carries real heterogeneity from one that carries almost none. The three pairs that fail to reject are informative precisely because they are so few, and they are not randomly placed: all three fall on sex or ethnicity, the two axes section 6 identifies as nearly homogeneous.

5. Result 3: the verdict follows the volume

To separate what the test measures from what the data contain, the same 2024–2025 loan-purpose table was counted at successively smaller scales, with every cell divided by a constant and rounded. The proportions are held;

only the sample size falls.

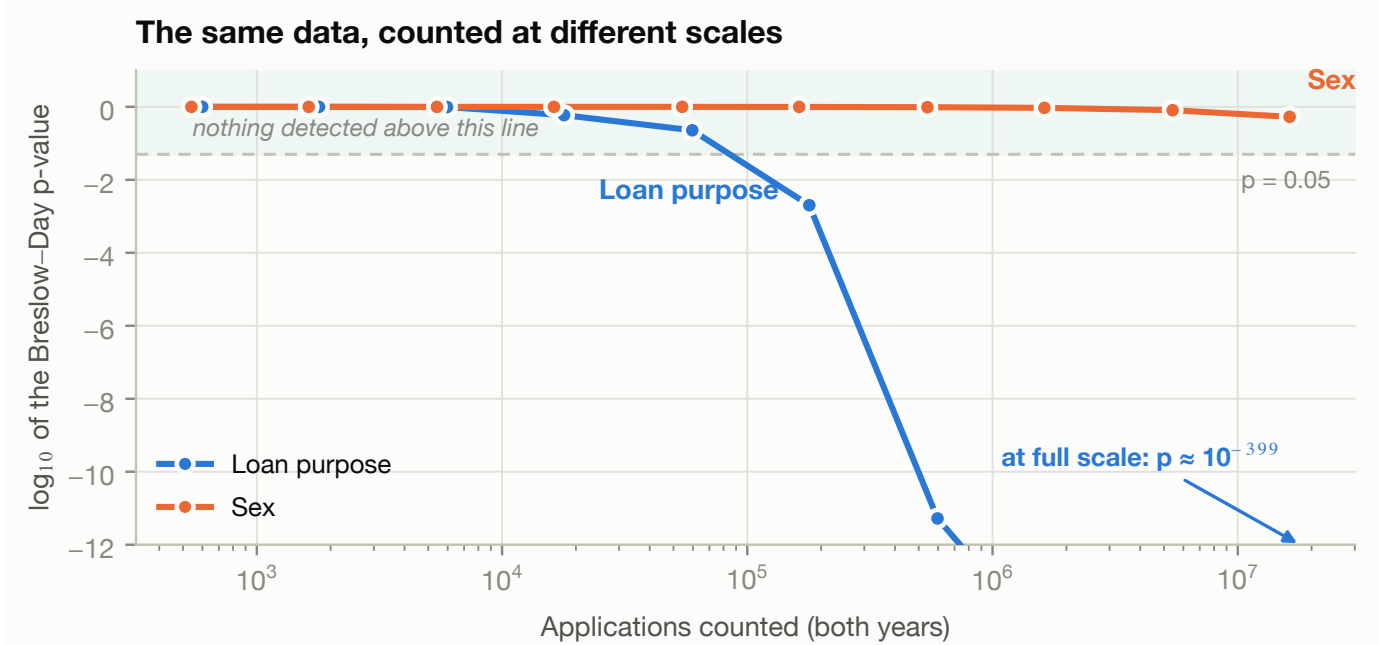


Figure 2. The Breslow–Day p-value for the 2024–2025 tables, recounted at ten scales. Cell counts are divided by a constant and rounded, so the proportions are held and only the sample size falls. Loan purpose crosses conventional significance near 10^5 applications and reaches $p \approx 10^{-399}$ at full scale; sex never crosses it. Below one hundredth of scale the per-cell rounding begins to distort the effect size, which is reported in Table 1.

Table 1. Identical proportions at five scales. Loan purpose against sex, applications for 2024 and 2025 combined. The p-value crosses conventional significance between the third and fourth row for loan purpose and never crosses it for sex.

Scale	Applications	Loan purpose, p	Loan purpose, spread	Sex, p	Sex, spread
1/10 000	1 793	0.999	1.18×	1.000	1.01×
1/1 000	17 895	0.731	1.14×	0.999	1.00×
1/100	178 951	0.002	1.79×	0.994	1.00×
1/10	1 789 515	10^{-37}	1.53×	0.938	1.00×
1/1	17 895 137	10^{-399}	1.53×	0.532	1.00×

At one thousandth of scale — roughly the annual volume of a mid-sized lender — the loan-purpose axis appears homogeneous at $p = 0.73$, while carrying an odds-ratio spread in the same band as at full scale. Two institutions with identical segment behaviour and different book sizes would reach opposite conclusions from the same test. Below one hundredth the per-cell rounding begins to distort the effect size itself, which is why the two smallest rows show spreads that drift; the p-value, by contrast, moves across four hundred orders of magnitude on proportions that never change.

THE PRACTICAL CONSEQUENCE

"Do our segments differ significantly?" is not a question about the portfolio. At the volumes a national aggregate carries, the answer is yes before the data are seen; at the volumes a single institution carries, it can be no for an axis that matters a great deal. The test does not tell an operations team which breakdowns to publish. The magnitude does.

6. Result 4: magnitude, and the order is not the intuitive one

Reported as the ratio of the largest to the smallest stratum odds ratio across the seven year-pairs, the eight axes separate by more than an order of magnitude, and the ranking does not follow the loan-versus-applicant division one might expect. Measured as excess over the neutral $1\times$, the widest axis carries 1.19 and the narrowest 0.08 — a factor of about fifteen.

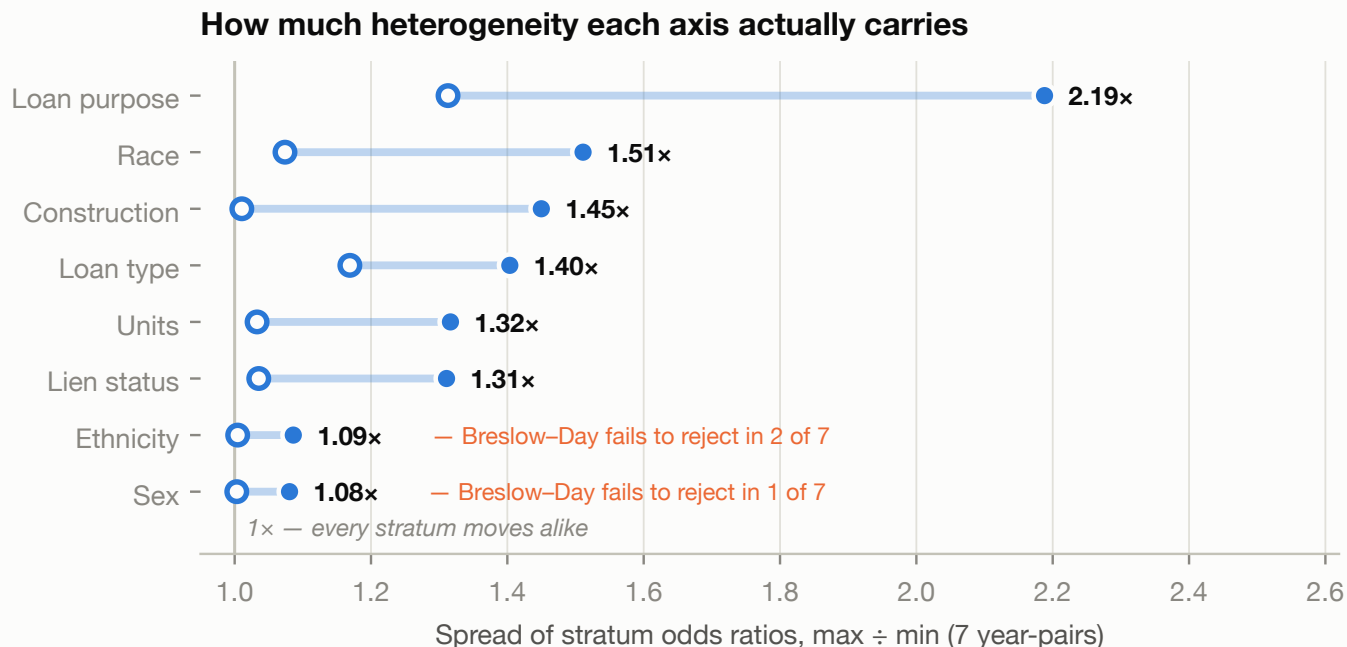


Figure 3. Heterogeneity carried by each segmentation axis, as the ratio of the largest to the smallest stratum odds ratio. The hollow marker is the narrowest of the seven year-pairs and the filled marker the widest; $1\times$ would mean every stratum moved alike. Sex and ethnicity are the only axes on which Breslow–Day ever fails to reject, and they are also the narrowest — an agreement between test and magnitude that does not hold for the other six.

Loan purpose is the widest at $2.19\times$, which is unsurprising: a cash-out refinancing and a home-improvement loan are different products assessed against different collateral. Race follows at $1.51\times$, ahead of loan type ($1.40\times$), the number of units ($1.32\times$) and lien status ($1.31\times$). Sex ($1.08\times$) and ethnicity ($1.09\times$) are effectively homogeneous, and they are the only axes on which Breslow–Day ever fails to reject.

6.1 Levels and movements are different questions

This result concerns year-over-year *movements* in approval rates, and must not be read as a statement about *levels*. The levels differ substantially. In 2025 the approval rate ranges from 64.90 % for applicants recorded as Native Hawaiian or Other Pacific Islander to 79.99 % for applicants recorded as White — a spread of 15.1 pp — and from 71.75 % to 81.00 % across the ethnicity categories, a spread of 9.2 pp. Those gaps are large, they are extensively documented in the fair-lending literature, and nothing here speaks to their causes.

Levels differ. The year-over-year movement does not.

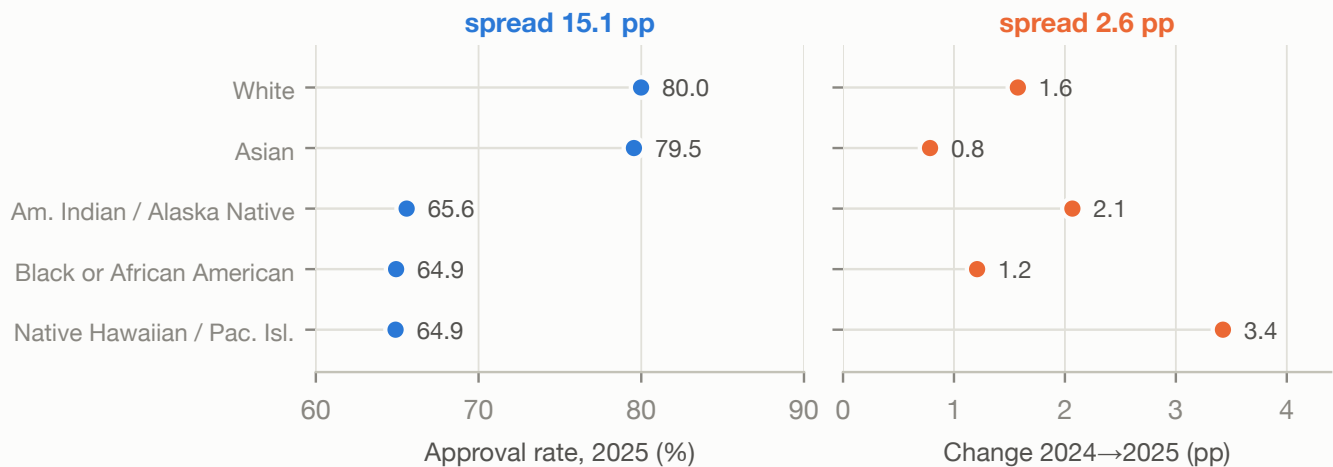


Figure 4. Levels and movements are different questions. Left: approval rate by reported race in 2025, spanning 15.1 pp. Right: the change in that rate between 2024 and 2025, spanning 2.6 pp. The homogeneity result of section 6 concerns the right panel only. Ethnicity behaves the same way, with a level spread of 9.2 pp and a movement spread of 0.5 pp.

What the homogeneity result says is narrower: between 2024 and 2025 those groups' approval rates all moved in the same direction by similar amounts, a spread of 2.6 pp against a level spread of 15.1 pp. A pooled year-over-year movement is therefore a fair summary of the sex and ethnicity strata and a poor one of the loan-purpose strata. A pooled *level* remains a poor summary of all of them.

6.2 What the spread costs the reader

Excluding the eight pairs whose pooled movement is smaller than 0.5 pp, where the ratio is an artefact of a near-zero denominator, the spread between strata is a median 0.65 times the pooled movement and exceeds it outright in 17 of 48 pairs. In other words, a reader given only the aggregate is routinely told a number smaller than the disagreement it conceals.

7. Discussion: publish the spread, not the test

Three claims follow, in decreasing order of confidence.

First, the Simpson warning should be retired from the standard briefing on approval rates, or at least demoted. It describes something that did not happen once in 56 opportunities across seven years of national data. Analysts taught to fear it are watching the wrong failure mode.

Second, a significance test on a heterogeneity question should not be reported by an institution whose volume it is a function of. This is not an argument against Breslow–Day, which is doing exactly what it was designed to do. It is an argument against using its output as a decision rule for which breakdowns deserve a dashboard tile. The alternative is direct: publish the spread of stratum odds ratios, or the range of stratum rate changes, both of which are volume-free and both of which appear in every table here.

Third, the choice of segmentation axis should be made on magnitude rather than convention. On these data, breaking an approval rate down by applicant sex adds essentially nothing to a pooled movement, while breaking it down by loan purpose changes the story entirely. Neither fact is available from a p-value, and one of them is the opposite of what a fair-lending dashboard's layout would suggest — which is a statement about measurement, not about which breakdowns an institution is obliged to monitor.

8. Limitations

The population is mortgage applications in 50 states and the District of Columbia, an unusually well-instrumented and unusually collateralised segment. Nothing here establishes that composition effects are equally negligible in unsecured lending, in credit cards, or outside the United States. The absence of Simpson reversals is a finding about this population and these eight axes, not a theorem.

The aggregated endpoint constrains the analysis to axes the API exposes one at a time. Interactions — a reversal visible only on purpose crossed with lien status, say — are not reachable this way and would require the loan-level files. Finer credit strata that plausibly carry more heterogeneity, notably binned income, loan-to-value and debt-to-income, were not examined, and neither was the lender level; a reversal may well exist at institution scale. Testing thousands of lenders for one would raise the multiplicity problem this paper's method is built to avoid, and any such search should be pre-registered.

The approval-rate definition excludes withdrawn and incomplete files. Including them lowers the level materially, since withdrawals are numerous; it does not affect the direction of any result reported here, but any comparison against a differently-defined published figure will disagree on level. The demographic fields are as reported by the filing institution, with substantial non-response, and the strata used are the API's own categories. HMDA filings are revised; the vintage used is stated in section 9.

Finally, the scale demonstration in section 5 divides and rounds cell counts rather than resampling applications. Rounding is why the effect size drifts in the two smallest rows of Table 1. A bootstrap would smooth that; it would not change the four hundred orders of magnitude the p-value travels.

9. Reproducibility

All series, analysis code and figure generation are available at github.com/cperezrisquet/approval-rate-reversal. The 64 API responses that feed the analysis are committed with it — they are public aggregate counts of roughly 500 bytes each — so every number here can be rechecked without contacting the API. The Breslow–Day p-values below the floating-point floor are computed from the asymptotic expansion of the upper incomplete gamma function, and the implementation is checked against `scipy` over the range where `scipy` does not underflow. The Mantel–Haenszel estimator is checked against `statsmodels`. Data retrieved 10 September 2026, covering HMDA filing years 2018 through 2025.

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Prepared from public sources. HMDA data is published by the Federal Financial Institutions Examination Council and is cited here as a data provider; nothing in this paper should be read as a criticism of its statistical practice, which documents the definitions the paper relies on, nor as a finding about lending discrimination — section 6.1 states what the homogeneity results do and do not cover. Filings are subject to revision by the source. · Dr. Sc. Carlos Pérez Risquet, September 2026